

Course-End Project Writeup

Crafting an AI-Powered HR Assistant: A Use Case for Nestlé's HR Policy Documents

Program: IITK AIML – Advanced Generative AI (Course 6)

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1. Situation

Nestlé's HR policies are documented in long PDF manuals. Employees and HR staff often spend significant time searching these documents for answers on leave, recruitment, working hours, and related topics. There is a clear need for a faster, self-service way to query policy content.

2. Task

Build a conversational AI assistant that answers Nestlé HR policy questions using Retrieval-Augmented Generation (RAG). The solution must load the Nestlé HR policy PDF, retrieve relevant passages, generate grounded answers with OpenAI GPT-3.5 Turbo, and expose an interactive Gradio chat UI.

3. Action

- **Environment setup:** Configured OpenAI API access and installed LangChain, ChromaDB, PyPDF, and Gradio.
- **Document ingestion:** Loaded the `_nestle_hr_policy_pdf_2012.pdf` with PyPDFLoader (8 pages) and split text using RecursiveCharacterTextSplitter (chunk size 1000, overlap 200) into 20 chunks.
- **Vector store:** Generated OpenAI embeddings and persisted chunks in ChromaDB for semantic retrieval.
- **QA system:** Built a RAG chain with GPT-3.5 Turbo and a prompt that restricts answers to retrieved context.
- **UI deployment:** Launched a Gradio ChatInterface so users can ask HR policy questions conversationally.

4. Result

The assistant retrieves relevant Nestlé HR policy sections and returns grounded answers. Example queries covered working hours / flexible work, recruitment / equal opportunity, and related HR topics. This demonstrates a practical enterprise GenAI pattern: PDF → embeddings → vector search → LLM → Gradio UI.

5. Tools & stack

Python · Jupyter Notebook · LangChain · OpenAI Embeddings · ChromaDB · GPT-3.5 Turbo · Gradio · PyPDF

6. Deliverables uploaded

- Writeup (this PDF)
- Screenshots of notebook pipeline, sample Q&A, and Gradio UI
- Source code notebook (`hr_assistant.ipynb`)

Note: Answers are generated only from retrieved Nestlé HR policy context to reduce hallucination.